

# Stiffness in the Robertson Chemical Kinetics System

What explicit methods pay — what implicit methods buy

Team of 3

Numerical Analysis of ODEs and PDEs — Project 1

September 24, 2026

One question: *how does stiffness translate into a practical step-size restriction for explicit methods — and what does an implicit method pay for the same accuracy?*

## The problem: Robertson chemical kinetics

$$\begin{aligned}y_1' &= -0.04 y_1 + 10^4 y_2 y_3 \\y_2' &= 0.04 y_1 - 10^4 y_2 y_3 - 3 \times 10^7 y_2^2 \\y_3' &= 3 \times 10^7 y_2^2\end{aligned} \quad y(0) = [1, 0, 0]^T, \quad 0 \leq t \leq 40$$

- ▶ Rate constants span  $10^8$ ; the state spans  $10^5$  ( $y_2 \sim 10^{-5}$  vs  $y_1 \sim 1$ )
- ▶ Invariant plane  $y_1 + y_2 + y_3 = 1 \Rightarrow$  structural zero eigenvalue
- ▶ Stiffness ratio over the two nonzero eigenvalues grows to  $1.584 \times 10^5$  at  $t = 40$

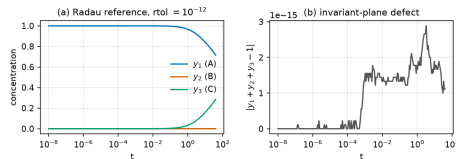
**Takeaway:** a benchmark designed so that explicit methods fail for a measurable, quantitative reason.

# What we built — and how we trust it

**From scratch:** explicit Euler, RK4, implicit Euler + Newton, adaptive step-doubling RK4 — swappable  $f$ , built-in cost counters.

**Our own oracle:** scipy Radau,  $\text{rtol} = 10^{-12}$ , tightened  $\sim 45\times$ ; digits that survive the repeat are the only ones we quote.

**Takeaway:**  $y(40) = (0.7158271, 9.1855 \times 10^{-6}, 0.2841637)$  to 15 digits; invariant defect  $2.9 \times 10^{-15}$ ; Problem Pack check value reproduced.



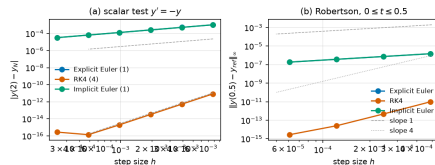
# Orders are verified, not assumed

Observed order (slope of  $\log err$  vs  $\log h$ ) on the stable window  $0 \leq t \leq 0.5$ :

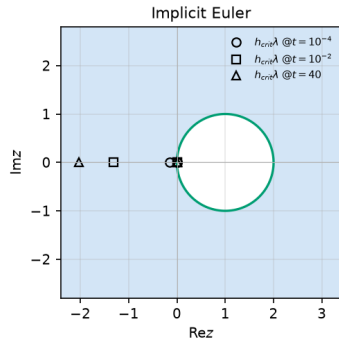
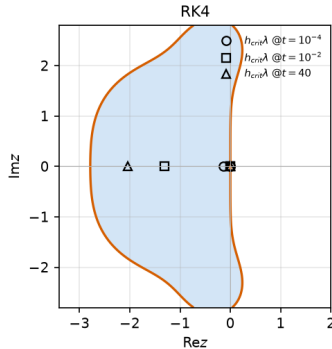
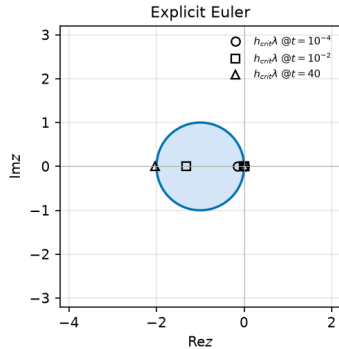
|           | Euler | RK4  | Impl. |
|-----------|-------|------|-------|
| scalar    | 1.00  | 4*   | 1.00  |
| Robertson | 1.00  | 4.35 | 1.00  |

\* round-off floor flattens the tail.

**Takeaway:** Euler and implicit Euler have the same order-1 constant — they differ in stability, not per-step accuracy.



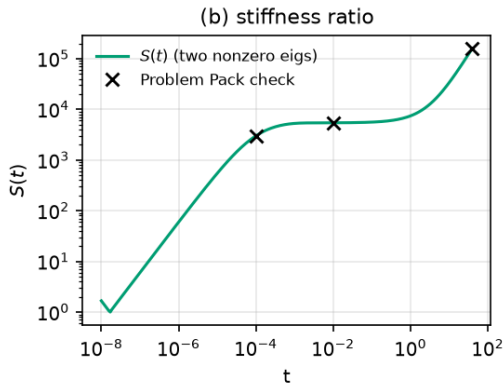
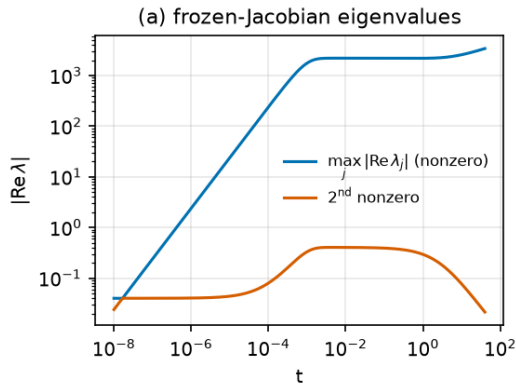
# Stability regions: hard ceilings for explicit methods



$h_{\text{Euler}} < 2/|\lambda|_{\text{max}} = 0.00059$ ,  $h_{\text{RK4}} < 2.8/|\lambda|_{\text{max}} = 0.00083$ , implicit Euler A-stable.

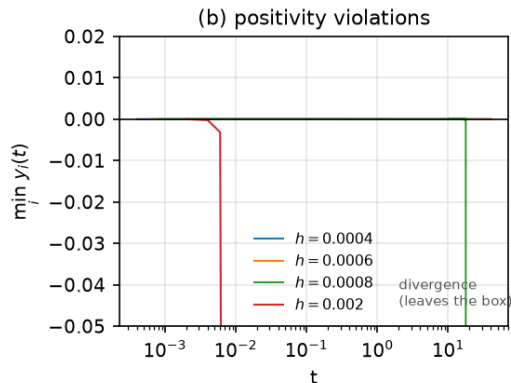
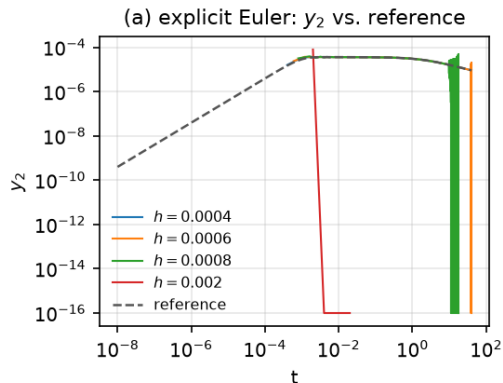
**Takeaway:** the frozen-Jacobian eigenvalue at  $t = 40$  sets a hard step ceiling — and we test it empirically next.

# Stiffness grows along the trajectory



**Takeaway:**  $S(t)$ :  $3.03 \times 10^3 \rightarrow 5.42 \times 10^3 \rightarrow 1.584 \times 10^5$  at  $t = 10^{-4}, 10^{-2}, 40$  — matches the Problem Pack check values.

# Explicit failure: physics fails before arithmetic



measured threshold  $h_{\text{crit}} = 0.0006$  vs. prediction 0.00059 (2% error).

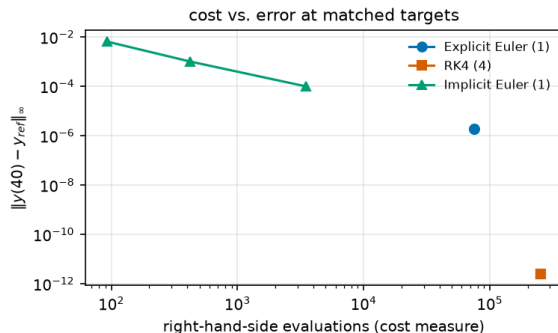
**Takeaway:** positivity is lost (bounded but non-physical) *before* blow-up — “did not explode” is not “is right”.

## Cost vs. accuracy at matched error

| Method         | Target | $h$     | error                 | nfev    | wall                  |
|----------------|--------|---------|-----------------------|---------|-----------------------|
| Explicit Euler | $10^2$ | 0.00053 | $1.9 \times 10^{-6}$  | 75,395  | 0.50 s (stab.-pinned) |
| Explicit Euler | $10^3$ | 0.00053 | $1.9 \times 10^{-6}$  | 75,395  | 0.51 s (stab.-pinned) |
| Explicit Euler | $10^4$ | 0.00053 | $1.9 \times 10^{-6}$  | 75,395  | 0.47 s (stab.-pinned) |
| RK4            | $10^2$ | 0.00063 | $2.6 \times 10^{-12}$ | 252,060 | 2.11 s (stab.-pinned) |
| RK4            | $10^3$ | 0.00063 | $2.6 \times 10^{-12}$ | 252,060 | 2.50 s (stab.-pinned) |
| RK4            | $10^4$ | 0.00063 | $2.6 \times 10^{-12}$ | 252,060 | 2.07 s (stab.-pinned) |
| Implicit Euler | $10^2$ | 2       | 0.0065                | 93      | 0.01 s                |
| Implicit Euler | $10^3$ | 0.3     | 0.001                 | 421     | 0.03 s                |
| Implicit Euler | $10^4$ | 0.029   | 0.0001                | 3,505   | 0.22 s                |

**Takeaway:** implicit Euler wins by  $810 \times / 179 \times / 21 \times$  at targets  $10^{-2}/10^{-3}/10^{-4}$  — explicit methods are pinned at their stability limit and cannot loosen up.

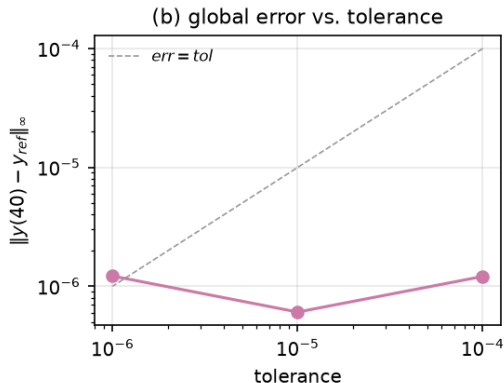
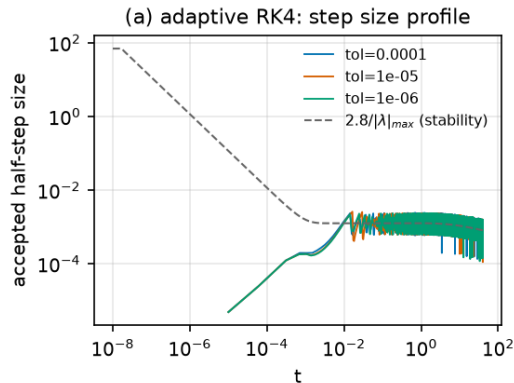
# RK4's order is wasted on the stiff interval



- ▶ RK4: error  $2.6 \times 10^{-12}$  *always* — its  $h$  is pinned at the stability limit for every target
- ▶ highest nfev, accuracy far beyond anything requested

**Takeaway:** on a stiff interval the binding constraint is stability, not accuracy — order alone does not win.

## Adaptive RK4: accuracy responds — stability does not



step size adapts over 3 orders of magnitude; controller tracks the  $2.8/|\lambda|_{\max}$  envelope from below.

**Takeaway:** error follows the tolerance down to  $\sim 10^{-6}$  and then *plateaus*: adaptivity buys accuracy, not stability.

# Conclusions

1. frozen-Jacobian prediction verified to  $\sim 2\%$  (0.0006 vs 0.00059)
2. at matched error, implicit Euler wins by  $21\times-810\times$  (nfev) on  $[0, 40]$
3. RK4's order 4 is irrelevant here — stability pins its step
4. adaptive controller: error plateaus at the stability ceiling
5. diagnostics fail in order: positivity  $\rightarrow$  stability  $\rightarrow$  (conservation survives)

**Takeaway:** stiffness is a stability story — measure it, don't assume it.

# Reproducibility

- ▶ `python code/run_all.py` regenerates every figure and every number (`results/results.json` → `numbers.tex` → report)
- ▶ `requirements.txt`: `numpy`, `scipy`, `matplotlib` — no other dependencies
- ▶ AI use documented in the AI Transparency Log (report App. B); ICS backed by git history (App. A)

Repository: `github.com/.../robertson-stiff-ode` — all team members can explain and modify every part (viva-ready).